

Chapter 8

University-Based Startups and Entrepreneurship: A Practical Guide for Industry Collaboration



Steve Gilison and Andy Wilson

8.1 Introduction

The Alliance for Southern California Innovation (the “Alliance”), a not-for-profit formed in 2017, has successfully brought together the heft of Southern California’s top research institutions, local business leaders, and world-class advisors to focus on bridging critical gaps in the SoCal innovation ecosystem. The goal of the Alliance is to engage and unify Southern California’s (SoCal’s) compelling diversity of talent, ideas, and perspectives in order to optimize the conditions for the region’s innovators to bring breakthroughs to the world. Alliance programs focus on developing and connecting SoCal’s diverse innovation communities while attracting new capital to Southern California by identifying, nurturing, and matching high potential startups to leading venture investors. By working together with community leaders, the Alliance tackles regional issues, including retaining top STEM talent and growing the base of local institutional capital. Andy Wilson, as the former founding Executive Director and CEO of the Alliance, focused on the organization’s Corporate Innovation programs. Steve Gilison, also a founding member of the team, serves as the Chief Operating Officer and leads the Alliance’s programs for Academic & Research Institutions.

Drawing from our experience in the Alliance, our chapter examines startups and entrepreneurship as an avenue for University-Industry (UI) engagement. Our goal is to provide actionable insights into programs and collaborations that enable startups and Intellectual Property (IP)-based startups (“spin-outs”) to make the leap from the lab to market and/or stimulate the creation of new startups or entrepreneurial solutions from students and faculty.

S. Gilison (✉) · A. Wilson

Alliance for Southern California Innovation, Pasadena, CA, USA

e-mail: sgilison@alliancesocal.org; Andy.Wilson@med.usc.edu

There is no shortage of sensational anecdotal tales featuring the inspired college student who dreams up a new idea in their dorm room, presumably while heating a cup of instant ramen, and then drops out to build that dream into a massively successful startup. Those stories are powerful in part because they appeal to our ideals of individual exceptionalism and are thus firmly embedded in our collective narrative about innovation in general. Yet the reality is that universities produce a considerable number of startups and “spin-outs” (when they start as research projects before leaping into the “real world”) whose paths to market success are long, arduous, collective efforts that rely more heavily on perseverance and injections of timely market-based input that builds on the core science, rather than individual brilliance and inspiration. We should not conflate the wildly different scenarios of clever undergrads creating the next Facebook in their dorms with that of a motivated post doc toiling in their lab working on life-saving cures for diseases or planet-saving decarbonization technologies, for example. In fact, the odds are stacked against the university-based startup so much so that the successful ones are the true “Uni-corns” (pardon the pun, very much intended).

While it is tempting to sell the promise of the anecdotal success story as a model of what could be, we have chosen to provide a more realistic and pragmatic overview of how university-based startups and spin-outs are created and the most common and useful engagement points for members of industry in that lifecycle. Our intention is to provide relevant frameworks and specific examples we’ve gathered from our own experience and from interviews we conducted with technology transfer and innovation representatives within universities and their corporate counterparts who invest resources into campus innovation broadly.

The main purpose of this chapter is to explain some of the current barriers to more productive UI collaborations that generate and accelerate new startups and then to describe the programmatic frameworks and examples that yield productive collaborations where industry is most effective at supporting innovative startups on campus.

8.2 Approach

For the past eight years we have had the pleasure of engaging with the universities in our region to support their innovation agendas and so we have a solid foundation of experience around the lab to market journey. However, since we do not directly work in academia, we thought it prudent to prepare for this chapter by gathering input via live interviews, group discussions, and written surveys from over two dozen innovation leaders at major universities and research institutions, mostly based in Southern California, of which nearly all have significant research budgets supporting world-class faculty. In addition, we spoke to several corporate innovation leaders who regularly engage with universities for collaborations that include IP licensing, entrepreneur mentorship, and other innovation-based programs.

Our interviews focused on three lines of inquiry:

- What are the most common and most valuable collaborations between industry and universities that drive startup formation and support?
- What are the most effective mechanisms (programs, tactics, etc.) that universities employ to develop meaningful industry partnerships that yield high value “investment” or collaboration that generate new startups or enable startups to successfully navigate from the lab to the market?
- Is corporate sponsored research a viable conduit to spurring small business creation (i.e., spin-outs) and entrepreneurship? Why or why not?

Our findings are distilled below for your benefit and we hope that this provides both actionable insights and inspiration for additional exploration.

8.3 Challenges and Opportunities

In this section, we identify and explain some of the well-established structural and market-based forces such as IP ownership and risk (in)tolerance which naturally limit the ability for Industry to directly drive startup creation and provide meaningful support (e.g., investment) to spin-outs.

8.3.1 *Sponsored Research and the IP Ownership Poison Pill*

When we first planned this chapter, we hypothesized that some small percentage of the billions of dollars (\$5.7B from for-profit sources and \$97.8B from all sources in FY 2022; NSF, 2023) spent annually by for-profit businesses on sponsored research in universities would lead to a new venture once the research was found to be promising but still too early to bring inside the sponsoring company, even if that venture is largely controlled by a corporate sponsor through governance and investment. This path undoubtedly spawns spin-outs in some rare cases but apparently it is so rare we did not uncover specific examples in our research. It was like tracking a Yeti, where people have heard they exist but never witnessed one firsthand in the wild.

There is a good reason for this scarcity. The goal of the sponsor (i.e., a corporation) is to utilize the research for proprietary purposes, such as de-risk some aspect of their R&D, develop an incremental enhancement to an existing product line or gain insights into new technologies that could disrupt their products. Because the university will almost always preserve their right to publish the findings, the sponsor is careful about their existing IP getting into the mix and thus the scope of the research is a piece of the puzzle and not a foundational technology. Whatever the end goal for the research, they typically require that the resulting IP will not be used by a competitor now or in the future. As such, the licensing agreements with the university will protect their investment and collaboration by giving them exclusive rights to the usage or commercialization of that IP and at least some guarantee of

governance in a spin-out. The primary motivation is for the IP to feed into the sponsor's R&D pipeline, not to exist outside of it.

This tie-up of the resultant IP makes it difficult for a spin-out to be nimble and innovate since there is now a significant limitation in how it operates such as restrictions on who can buy or sell their products. The restrictions also limit the spin-out's potential sources of operating capital since outside investors see that IP restriction (and cap table if the sponsor is an equity holder) as a red flag. Without access to capital markets, the spin-out is reliant on the original sponsor organization for its funding while it scales to reach profitability or an exit. Since corporations have a tendency to change priorities via annual budgets, new leadership with fresh mandates, and success metrics that don't align well to startup lifecycles, these types of investments and projects often get snuffed out before they have the chance to succeed and become self-reliant.

From the research faculty's side, the encumbrance on the IP is essentially a poison pill that makes a successful exit unlikely, thus hampering their motivation to make the difficult leap to entrepreneurship. And from their perspective, life is good: their research is getting funded, they are generally allowed to publish their findings (often a motivation of the sponsor to validate their product) and their research has a fighting chance to make its way into the world due to the collaboration with industry—all of this without the faculty taking any startup risk. Simply put, the incentives are not aligned for spin-out creation and therefore it makes sense that this is not a robust pathway for startup success. If you do see one succeed in the market despite these structural barriers, please take a non-grainy photo with ample light and send it to us (not the tabloids).

8.3.2 Spin-Outs' Risk Profile and Lifecycle Stage Substantially Curbs Industry Appetite for Investment

For the past seven years, we have produced the First Look SoCal showcase, which is an annual showcase of frontier technology and life science spin-outs from the top research institutions and universities in the Southern California region (e.g., University of California at Los Angeles (UCLA), University of Southern California (USC), California Institute of Technology (Caltech), University of California at San Diego (UCSD), and over a dozen others). All of the 24 startups selected each year to pitch to a room full of investors and corporate leaders are looking for their first round of private investment. Almost all of them have received either some form of non-dilutive funding through their universities (e.g., "Proof of Product" grants) or public/government grants (NIH, NSF, DOE, DOD, SBIR/STTR, etc.) as well as possess IP protected by their technology transfer offices whose due diligence justified the investment in patents and legal fees. Despite the high quality of the underlying IP and potential market need, very few will raise enough funding from private investment to be successful and of those who do, almost none of them will get early

stage funding from a corporation or their Venture Capital (VC) funds. This is not a SoCal thing—this is generally true for spin-outs across the country, for a number of reasons.

Why don't corporations provide funding or other forms of meaningful support to spin-outs? The short answer is that they invest in later stage startups with more traction and less risk. The longer answer is more interesting. Generally, corporations who invest in startups do so through an in-house VC fund, i.e., "CVC," which usually applies both strategic and financial criteria to make investment decisions. The CVC's we spoke with strategically invest in startups where they can gain valuable insights, or the startups' success will help the market or the corporation's customers. Even if the startup is strategically interesting, they will only invest if the startup has the right team and market strategy where the funding will help them thrive and scale, not just get to the next early development milestone. The subset of corporations who fund pre-seed rounds will usually want the startup to have an experienced team with a track record of success—not the typical profile of a university-based spin-out! We could write another chapter about the need and challenges of aligning PI's with experienced operators but we'll save that for the next volume.

Many corporations will also partner with startups to conduct pilot programs to assess technical and market validation, but the startups need to have a sufficient "Technical Readiness Level" (TRL) to justify the risk as well as the investment in time and resources to run the pilot. However, most spin-outs and university-based startups are not at a sufficient TRL to qualify for a corporate pilot. This lifecycle gap is a bit of a "Catch-22," because startups typically look to increase their TRL with validation of a corporate pilot program as a step toward funding.

Similarly, traditional VC funds also employ financial criteria that drive investment in experienced teams with validated technology and are pursuing large market opportunities so the investor can justify the risk of their investment while knowing that at least 75% of their investments fail (Pullman, 2023). The more well-funded ones will sponsor research on behalf of their portfolio companies which sometimes include research projects they are currently incubating. The relatively small number of VC firms who will invest in spin-outs typically have much smaller funds to deploy but take a hands-on approach by providing value-added technical and market domain expertise. These VC firms are invaluable but far too few exist and they tend to focus on narrow industries that align with their expertise. Deep domain knowledge is a prerequisite for these specialty firms as they often come in at a stage where the science is evolving simultaneously with early exploration of commercial market viability. A good example of a technology-transfer focused boutique VC firm is [Freeflow Ventures](#), which is currently investing from a second fund (total AUM = \$90m) and has invested in over 30 spin-outs almost exclusively from Caltech. Another source of spin-out funding comes from experienced angel investors who bring relevant market experience and funding to help build out early proof of concept milestones and customer discovery initiatives. While angels play an important role, especially those with years of relevant industry expertise, most invest part-time, don't aggressively source deal flow and only fund deals in their local geographies which makes finding a good match at the right time often difficult.

Arguably, the scarcity of “smart” value-add money in the system naturally limits the number of successful spin-outs. But let’s also be realistic: more firms and larger funds would exist if spin-outs represented a compelling investment opportunity. Early stage boutique VC firms and even some angels are susceptible to the same business model constraints as traditional VCs: one successful startup funds the nine others who fail or just break even. So unless the likelihood of return on investment increases through de-risking strategies (e.g., industry collaboration, more government/non-dilutive investment, higher quality mentorship, etc.), the supply of available capital will remain below demand levels. As the fundraising mantra goes: “Ask for money and they’ll give you advice. Ask for advice and they’ll give you money.” Given the challenge of funding these early-stage spin-outs, most founders receive a lot of advice. Nonetheless it is common to find VCs, incubators, and venture studios roaming the halls of universities to stay close to break-through innovations and top talent, even if not for the deal flow.

Lastly, it is important to note several corporations have launched their own incubators where they fund and support the development of university research projects into spin-outs that typically align with their product portfolios. This mechanism seems to be concentrated in biotech and especially biopharma such as Lilly Ventures from Eli Lilly and JLABs from Johnson & Johnson. For biopharmaceuticals, it makes a lot of sense to choose high potential projects that might fit into their product portfolio and then use a hands-on incubation model to boost the development lifecycle while locking in the future value capture for the parent corporation. This is also an excellent option for biopharma researchers who have very few paths to commercialization and will benefit from tapping into the vast resources of these companies. This model seems like it should work with other industries where the scientific breakthrough is very close to the end product but less likely to exist where the science is well upstream from market-ready products.

8.4 Options, Solutions, and Impact

Despite the numerous structural impediments, a number of university-led programs engage industry to gain their invaluable market-based insights, which then spur meaningful innovation and entrepreneurship. Several are outlined here. We also describe emerging opportunities that could be meaningful sources of innovation and impact globally if more widely adopted.

8.4.1 Dynamics that Drive Spin-Out Creation and Successful Industry Engagement

We could fill an entire chapter, even a book, on the challenges faced by spin-outs on their path from lab to market but since that is already well documented we'll spend just a moment to cover what is specifically relevant to UI collaboration in this uniquely challenging journey. First and foremost, spin-outs fail due to a lack of market-driven value creation—the technology might be impeccable but too often does not fill a market need that is large enough to merit the high risk of investment needed to fund the lengthy buildout of early stage innovations emerging from the lab. While some are ineffectively building solutions looking for a problem (or too small of a problem), many are building innovative solutions with a valid market application for known real-world problems with sufficient commercial opportunity. However, the founder is typically a scientist who does not have sufficient access to “voice of the market” input so they run head first into a wall of market-based impediments that experienced industry professionals might have anticipated such as the presence of deeply entrenched competition or inhospitable distribution channels, unattractive costs of manufacturing at scale, or long sales cycles to a small number of potential customers, to name just a few. Ideally, the scientific founder finds (or is matched with) an experienced business entrepreneur or in some cases an EIR (entrepreneur in residence) with sufficient domain expertise to lead the non-scientific aspects of the startup including raising institutional investment to tilt the odds of success. Marx and Hsu (2022) explain as follows: “we find that teams of academic scientists whose former collaborators include ‘star’ serial entrepreneurs are much more likely to commercialize their own discoveries via startups.”

Spin-outs who don't immediately raise funding while the founders are still employed by the university still receive meaningful support from their university “innovation” functions, usually within the technology transfer office. Most large universities have dedicated departments or groups within the department to support spin-outs who work alongside the technology transfer function such as the New Ventures Group within the Technology Development Group at UCLA or purpose-built centers like the Austin Technology Incubator whose primary function is to support promising faculty-based spin-outs. Over the past eight years, we have seen a marked increase in the amount of resources invested by universities in building out their capacity to support new venture creation and entrepreneurship; core to this investment is expanding their ability to get industry involvement to support the spin-outs and entrepreneurs with mentorship and guidance beyond the typical sponsored research collaborations and IP licensing deals that historically dominate the scope of their partnerships. More recently we are seeing a trend of retaining EIRs (executives in residence) either on a full time basis to support multiple spin-outs or alternatively a “cast” of part time EIRs with diverse experience. This type of talent investment to support entrepreneurially minded PIs appears to be an effective strategy to help spin-outs cross the chasm from lab to market.

We were told (more than once) that the sure way to kill truly disruptive innovation is to have industry directly initiate startup and spin-out creation at universities. Industry is typically more valued when they are brought in at the right time by the startup or the university group (e.g., on campus incubator, technology transfer office, etc.) as they seek to better understand customer needs and prioritize market opportunities. It is still valuable for industry to initiate competitions focused on challenge statements and problem/solution hackathons to engage students and faculty to help them solve real-world problems—the participants get a lot out of the experience and they often inspire or inform future entrepreneurship and innovation in other areas. It also provides an effective way for corporations (talent consumers) to evaluate and start engaging students that align with future talent needs. However these activities are not especially effective at spurring startup creation.

The most common refrain we heard from university leaders is that in general, each startup and spin-out is unique so a one-size-fits-all approach for industry input is not effective. The university (or relevant support program) brings in the industry as needed and when the timing is right. Luckily, the universities get access to a lot of industry members through other areas of engagement, typically through partnerships around sponsored research, talent recruitment programs, and philanthropic support. In fact, it is often an expectation (and sometimes a requirement) of the partnership agreement that industry will lend their time and expertise to assist the entrepreneurial programs even if they are providing material support in other areas of the university. Corporations need the skilled workforce emerging from the universities so it makes sense to use the talent pool as leverage to negotiate their involvement in entrepreneurship and research validation programs where they have a uniquely valuable perspective to share. Many universities reap significant licensing revenue from their spin-outs so it makes sense to leverage institutional assets like human capital and the reputations of research faculty to compel industry to also lend their time (and money) to benefit higher quality spin-outs with improved likelihood of market success.

8.4.2 Proof of Concept Programs for Quality Outcomes

One of the key differentiating factors of a successful spin-out is getting high-quality input from industry early and often. Even though it is hard to quantify, getting market-based insights is the most important and prevalent high value input that industry can provide to spur innovation and startup success. One of the most productive and popular ways industry engages with early stage innovation is to support translational research teams by participating in proof of concept or “proof of product” incubator programs (“PoPs”). These often include university-funded non-dilutive grants ranging from \$10K to \$100K whose primary aim is to advance the translational research by hitting a defined benchmark that will reveal key insights about the technical and/or market viability of the project, typically lasting a few

months or up to a year. These programs combine mentorship, typically with subject matter experts from within the university and industry experts in combination with some entrepreneurial training to help validate the market-based components.

Industry members often review projects during the application phase to help the university decide which ones join the PoP program. They also mentor the teams throughout the program and give feedback at the end. At the end of the proof of product program, many of these promising innovations are shelved for one reason or another but those that remain have a strong foundation for justifying a decision to move forward with a spin-out. The net result is stronger spin-outs that have received vital input and preliminary validation from industry early on and a better chance at leaping from lab to market. Sometimes the industry organizations will continue to track and engage with the spin-out as they progress through their early stages and eventually, in theory, provide some support. This makes a lot of sense since they helped the research team focus their innovation efforts to address a market need they identified and are already in a productive relationship with the founding team when the timing is right for investment or other material support. This kind of long-range thinking on behalf of corporate R&D and business development departments is a good example of how university-based spin-outs benefit from industry collaboration.

Industry feedback loop mechanisms are also baked into government-led incubation programs like the NSF's I-Corps and SBIR/STTR programs that mandate industry representatives provide direct input in the project. The I-Corps program has launched more than 1000 startups since its inception in 2011 (NSF, 2024). The startups have cumulatively raised over \$760 million in subsequent funding. The program focuses on validating the business elements along with the technical and includes members of industry as integral inputs to the success of the spin-outs that enroll in the program. The bedrock of the I-Corps is teaching entrepreneurship to academics rather than providing funding to advance science. NSF Director Sethuraman Panchanathan shared some reflections as a recent participant in the program: "Success is not always measured by whether you create an entrepreneurial venture. It is also measured by how you take the entrepreneurial mindset and apply it in all of the things you do in your life." This kind of entrepreneurship training is one of the many ways that industry helps infuse invaluable skills and knowledge into the university startup ecosystem.

8.4.3 Innovation Centers and Consortia

We learned of some university-led programs that take innovative approaches to engaging industry to partner with the innovation functions within the institution. These are mostly tweaks to PoP programs but with a very specific focus on the expectation of collaboration by the company, not just a "pay it forward" service provided by representatives of the company, as described above. While there are

definitely many more examples in the market, these two models might inspire other university leaders to explore new pathways with their faculty and industry partners.

Innovation centers at universities will often engage several corporations to provide market insights into a vertical area of interest so that they can inform and influence the startups emerging from their labs. The UCSD innovation center often starts with a focus area of interest, such as mobile technology (not theoretical, they have Qualcomm's HQ in their backyard) and will bring in multiple corporations to serve in an advisory capacity by providing insights into the future gaps and opportunities for technological solutions. Paul Roben, Associate Vice Chancellor, Innovation and Commercialization, at UCSD, explains it like this: "It's not that the corporate comes to us saying, I want you to start a company that I can acquire; it's us going to the corporate and saying, we want to run this program in your area of expertise and we want you to help us understand what the market is looking for." One of the side benefits of building coalitions of industry leaders to gather market insights prior to development is that the representatives, often from competing or partner organizations, will form relationships with each other and launch collaborations that incorporate solutions being developed by researchers in the PoP-like programs. This type of coalition is also employed when working with multiple Department of Defense organizations who each have specific mandates but share common challenges and have incentives to repurpose innovations across branches.

Another interesting example is the industry consortium model at University of California at Santa Barbara's Solid State Lighting & Energy Electronics Center (SSLEEC) that utilizes a membership model where corporations collectively fund the research and in turn they have preferential access to license the IP that is owned by the university. Corporate members frequently assign one of their internal experts to be a "visitor" at the center for up to a year to collaborate on the research and inject a wealth of market insight into the technologies being developed. The researchers work closely with industry to build innovations with a clear market need so their research is infused with a sophisticated understanding of the market dynamics, cost drivers, scalability, and viability to a degree not commonly found in most university labs. Even if these entrepreneurial researchers do not aspire to make the leap from lab to market themselves, their innovations certainly are doing so. And if they do decide to make that leap, as several have done at the SSLEEC—most notably Soraa Laser Diode (SLD Inc.), which was later acquired by Kyocera (2020)—there is technically nothing stopping them, since the startup is able to license the IP from the university like usual because the member corporations do not own the IP (avoids the aforementioned IP "poison pill"). It is conceivable that a consortium model focused on other technologies with potentially lower barriers to entry or different market dynamics could lead to a proliferation of startup activity and perhaps even the consortium itself collectively supporting a spin-out without the constraints of IP ownership by one of the members thwarting potential commercial availability. Dare we dream?

8.4.4 Limitations

We found considerable evidence that industry plays a vital role in the entrepreneurial output at universities, largely by providing “voice of the market” input through various university-led programs. There is a rather well-worn roadmap that most universities follow in one form or another to transform research projects into startups (i.e., spin-outs) and invariably those spin-outs will have significant interaction with industry, typically via mentorship and expert guidance. We consistently heard that valuable collaborations with industry rarely result in startup formation and that spin-outs seldom receive direct funding from the corporates or corporate VCs (CVCs) until after they’ve left the university’s orbit. While there are many beneficial interactions with industry in the university-based startup journey, they are rarely a catalyst or significant resource for tangible support like funding, IP, or go-to-market support (pilot programs, first customer, distribution channel, etc.). In fact, our interviews with innovation leaders at universities and corporations revealed scant recollection of any startups created via direct collaboration with a corporation.

Furthermore, we did not find a mother lode of startups emerging from a specific type of university-led program where the launch and/or success of the startup was due to industry collaboration. This is especially true of industry-sponsored research, which rarely results in startups, joint ventures, or IP licensing to a third party. Our conclusion is that a variety of well-established structural and market-based forces naturally limit the potential for startup creation and direct support via industry collaboration, yet there is ample opportunity for the collaborations to spur meaningful innovation and entrepreneurship broadly that has the potential to ultimately lead to startup creation. The value of these collaborations is usually expressed as direct monetary gain (e.g., research funding for the university or enhancements to existing products for industry) as well as experiential knowledge gathering that enhances both stakeholder organizations’ human capital and IP assets.

8.5 Takeaways and Future Strategies

It cannot be overstated that industry plays a massively important role in spurring innovation on campuses worldwide and that their input and support spurs startup creation. The most common and valued input is providing researchers and founders with priceless “voice of the market” information. Ideally that input is given at multiple inflection points in the lifecycle of the spin-out, from the research project or hackathon all the way through fundraising. Here is a representative list of entry points for industry input, listed from most nascent to least:

- Provide input and guidance to the university’s innovation leadership and faculty (usually technology transfer office or adjacent) about current and future technology needs and gaps in your sector
- Evaluate nascent research project proposals seeking proof of product/concept grants

- Mentoring research teams seeking grants such as SBIR/STTR grants where gathering voice of the customer is required condition of the application process
- Issuing challenges and instigating student hackathons that stimulate innovation through problem solving and entrepreneurship
- Invest in consortium/membership model research centers through funding and lending expertise to shape research; also be supportive of spin-outs that emerge from the center to accelerate the commercialization of developed innovations
- Judge on-campus startup showcases and competitions to provide vital input to nascent founders
- Invest in early stage startups emerging from campuses!!!

It has been extremely encouraging to see the breadth and depth of the infrastructure being built by the universities to engage industry during these key inflection points in their innovation capacity. Similarly, with the pace of innovation increasing, we have seen more and more industry players taking serious interest in university-based innovations and researchers as the potential source of the next generation of game changing breakthroughs.

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Steve Gilison is the COO at the Alliance for SoCal Innovation where he especially enjoys leading programs that support academic and research institutions with a focus on commercialization and entrepreneurship. With over two decades in tech, he has held leadership positions at startups and major companies like online publisher Internet Brands and the accounting firm Deloitte. Gilison holds a BA from the University of Maryland and an MBA from UCLA.

Andy Wilson is a seasoned executive with leadership experience across public, private, and non-profit sectors. He has co-founded nine startups, including two nonprofits. As an entrepreneur and venture capitalist, he has focused on innovation-driven startups, often leveraging university research. After 7.5 years building the Alliance for SoCal Innovation, he recently joined Keck Medicine of USC to orchestrate strategic innovation.